UPSC EPFO · APFC RECRUITMENT TEST · FOUNDATION LAYER · MODULE 9 · CHAPTER F9.7

Networking, the Internet and HTML

Chapter F9.7 — The Last Three 2023 Items, and the End of Module 9

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NOTE · THE LAST THREE 2023 ITEMS

This chapter closes the list. The final three of the twenty 2023 computer and science items are here: **IP address validity** (§4.2), **mesh topology links** (§2.2) and **HTML internal linking** (§6.3).

Two of the three are **deterministic**: the mesh-link count is a formula, and IP validity is a range check. Like the number-system conversions in F9.5, they cannot be got wrong once the rule is known — which makes them the best-value pages in the chapter.

Table of Contents

0. How This Chapter Works

STUDY SESSION · CHAPTER F9.7 — SIX SESSIONS

1. **Session 1** — Network types and transmission (§1) ★★★★☆
2. **Session 2** — Topologies, and the **mesh formula** (§2) ★★★★★ **2023**
3. **Session 3** — Devices and the OSI model (§3) ★★★★☆
4. **Session 4** — **IP addressing** (§4) ★★★★★ **2023**
5. **Session 5** — The internet and its protocols (§5) ★★★★☆
6. **Session 6** — **HTML**, including internal links (§6) ★★★★★ **2023**

Then 25 graded questions in §7. Budget **two and a half days**.

1. Session 1 — Network Types and Transmission ★★★★★

1.1 By geographical area

Network	Area covered
PAN — personal area network	A few metres, around one person
LAN — local area network	A **room, building or campus**
MAN — metropolitan area network	A **city**
WAN — wide area network	A **country or the world**. **The internet is the largest WAN**

Client-server networks have a central server providing resources; **peer-to-peer** networks treat all nodes as equals.

1.2 Transmission media and modes

Medium	Notes
Twisted pair	Cheap, common, limited bandwidth
Coaxial cable	Better shielding
Optical fibre	**Highest bandwidth**, **immune to electromagnetic interference**, no crosstalk
Unguided	Radio, microwave, infrared, satellite

Mode	Direction	Example
Simplex	**One way only**	Keyboard to computer, radio broadcast
Half duplex	Both ways, **one at a time**	**Walkie-talkie**
Full duplex	**Both ways simultaneously**	**Telephone**

MUST MASTER · OPTICAL FIBRE AND THE DUPLEX PAIR

Optical fibre carries the highest bandwidth and is **immune to electromagnetic interference**, because it transmits light rather than electricity. Both properties are asked, sometimes together.

And the duplex distinction: **half duplex allows both directions but only one at a time** — the walkie-talkie, where you must stop talking to listen. **Full duplex allows both at once** — the telephone. Options routinely swap the two examples.

CHECK YOURSELF · SESSION 1

1. Order PAN, LAN, MAN and WAN by area. Which is the internet?
2. Which medium has the highest bandwidth, and what interference is it immune to?
3. Distinguish simplex, half duplex and full duplex, with an example of each

2. Session 2 — Topologies ★★★★★

2.1 The four topologies

Topology	Structure	Strength	Weakness
Bus	All nodes on a single **backbone cable**	Cheap, simple	**Failure of the backbone kills the network**
Star	All nodes connect to a **central hub or switch**	Easy to add nodes; one node's failure is isolated	**Failure of the hub kills the network**
Ring	Each node joined to two others in a **closed loop**	Orderly data flow	A break can disable the ring
Mesh	**Every node connected to every other node**	**Most reliable and robust**	**Most expensive**; most cabling

2.2 The mesh formula — the 2023 item

MUST MASTER · THE NUMBER OF LINKS IN A FULL MESH

For a full mesh network of **n** nodes:

> **Number of links = n(n - 1) / 2 Number of input/output ports per device = n - 1**

The reasoning, so you never have to recall it blindly: each of the n nodes must connect to the other **n - 1** nodes, giving n(n - 1) connections — but every link has been counted **twice**, once from each end, so divide by **2**.

Worked values worth having ready:

Nodes (n)	Links
4	6
5	**10**
6	**15**
8	28
10	**45**

The distractor is always **n(n - 1)** without the division — so 30 instead of 15 for six nodes. If an option is exactly double another, the doubled one is the trap.

COMMON UPSC TRAP · WHICH TOPOLOGY IS "EVERY NODE TO EVERY NODE"

That description is **mesh**, and it is offered against bus, star and ring in almost every topology question.

Keep the four straight by their single point of failure: **bus fails at the backbone**, **star fails at the hub**, **ring fails at a break**, and **mesh has no single point of failure at all** — which is precisely why it is both the most reliable and the most expensive.

CHECK YOURSELF · SESSION 2

1. Name the four topologies and the single point of failure of each
2. Give the formula for links in a full mesh, and derive it in one sentence
3. How many links for 5 nodes? For 8? For 10?
4. How many ports does each device need in a full mesh?

3. Session 3 — Devices and the OSI Model ★★★★☆

3.1 Networking devices

Device	Function	OSI layer
Hub	**Broadcasts** to all ports; no filtering	**Physical**
Switch	Forwards using **MAC addresses**	**Data link**
Router	Connects **different networks** using **IP addresses**	**Network**
Bridge	Joins two LAN segments	Data link
Gateway	Connects **dissimilar** networks; converts protocols	Application
Repeater	**Regenerates** a weakened signal	Physical
Modem	**Mod**ulates and **dem**odulates — digital to analog and back	Physical
NIC	Network interface card; carries the **MAC address**	Data link

3.2 The OSI model

Seven layers, from the bottom up:

> **Physical » Data Link » Network » Transport » Session » Presentation » Application**

MUST MASTER · SEVEN LAYERS, AND WHICH DEVICE SITS WHERE

The count itself is asked: **OSI has SEVEN layers; TCP/IP has FOUR** (network access, internet, transport, application).

A mnemonic from the bottom up: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way.

And the device mapping, which is the more common question:

Device	Layer
Hub, repeater	**Physical** (layer 1)
Switch, bridge, NIC	**Data link** (layer 2)
Router	**Network** (layer 3)

A router works at the network layer using IP addresses; a switch at the data link layer using MAC addresses. That pair is the single most examinable distinction in the section.

TCP versus UDP:

	TCP	UDP
Connection	**Connection-oriented**	**Connectionless**
Reliability	**Reliable**, ordered, acknowledged	**Unreliable**, no guarantee
Speed	Slower	**Faster**
Used for	Web pages, email, file transfer	Streaming, voice, gaming

CHECK YOURSELF · SESSION 3

1. Which device broadcasts to all ports? Which forwards by MAC address? Which by IP address?
2. Name the seven OSI layers from the bottom up. How many layers has TCP/IP?
3. At which layers do a hub, a switch and a router operate?
4. Give four differences between TCP and UDP

4. Session 4 — IP Addressing ★★★★★

4.1 IPv4 and IPv6

	IPv4	IPv6
Length	**32 bits**	**128 bits**
Notation	**Dotted decimal** — four octets	**Eight groups of four hexadecimal digits**, colon-separated
Example	192.168.1.10	2001:0db8:...
Address space	About 4.3 billion (2^{32})	Vastly larger

4.2 Validity — the 2023 item

MUST MASTER · WHAT MAKES AN IPV4 ADDRESS VALID

An IPv4 address is valid only if it has **exactly four parts**, each part being a whole number **between 0 and 255 inclusive**.

Why 0 to 255? Each part is one **octet** — eight bits — and eight bits represent 2^8 = **256** distinct values, running **0 to 255**. The count is 256; the **maximum is 255**.

So the two ways an address fails, and both appear as options:

Address	Verdict	Reason
192.168.1.10	**Valid**	Four parts, all within range
256.100.50.25	**Invalid**	**256 exceeds the maximum of 255**
300.1.1.1	**Invalid**	300 far exceeds 255
192.168.1	**Invalid**	Only **three** parts

The off-by-one is the trap: **256 is not a valid octet value**, even though there are 256 possible values. An option offering the range "1 to 256" is wrong at both ends — it excludes 0 and admits 256.

Address classes: A (first octet 1 to 126), B (128 to 191), C (192 to 223), D (224 to 239, multicast), E (240 onwards, experimental). **127 is reserved for loopback**, of which 127.0.0.1 is the familiar case.

Private ranges, not routable on the internet: 10.x.x.x · 172.16 to 172.31.x.x · **192.168.x.x**.

A **MAC address** is different in kind: **48 bits**, written in hexadecimal, and **burnt into the hardware** rather than assigned by the network.

CHECK YOURSELF · SESSION 4

1. How many bits in IPv4? In IPv6? How is each written?
2. State the two conditions for a valid IPv4 address
3. Why is the maximum octet value 255 and not 256?
4. Which addresses are these — valid or not: 10.0.0.1, 192.168.300.1, 172.16.5, 255.255.255.0?
5. What is 127.0.0.1? How many bits is a MAC address?

5. Session 5 — The Internet and Its Protocols ★★★★☆

5.1 The web

The **World Wide Web** was invented by **Tim Berners-Lee**. The web is a service running **on** the internet; the internet is the underlying network — an examinable distinction.

A **URL** — Uniform Resource Locator — is the address of a resource. A **browser** renders pages; a **search engine** indexes them; an **ISP** provides connectivity.

5.2 The protocols

Protocol	Purpose	Port
HTTP	Transfers web pages	**80**
HTTPS	HTTP **secured** by encryption	**443**
FTP	**File** transfer	**21**
SMTP	**Sending** email	**25**
POP3 / IMAP	**Receiving** email	110 / 143
DNS	Translates **domain names into IP addresses**	53
DHCP	**Assigns IP addresses dynamically**	—
TELNET	Remote terminal access	23

MUST MASTER · SEND VERSUS RECEIVE, AND NAME VERSUS NUMBER

Two pairs that carry most of the protocol questions.

1. **SMTP SENDS mail. POP3 and IMAP RECEIVE it.** The mnemonic is in the name: SMTP is Simple Mail **Transfer** Protocol — transfer outwards. An option saying SMTP is used for receiving email is wrong, and it is the commonest error in the topic
2. **DNS translates names into addresses. DHCP assigns addresses.** DNS looks up; DHCP hands out. Both deal in IP addresses, which is exactly why they are confused

Port numbers worth knowing: **HTTP 80, HTTPS 443, FTP 21, SMTP 25, DNS 53**.

Security: a **virus** attaches to a host file; a **worm** spreads by itself; a **Trojan** disguises itself as legitimate software; **ransomware** encrypts data for payment; **phishing** deceives a user into disclosing credentials. A

firewall filters traffic between networks. **Encryption** protects content; a **digital signature** authenticates the sender.

Cloud computing is offered as **IaaS**, **PaaS** and **SaaS** — infrastructure, platform and software as a service.

CHECK YOURSELF · SESSION 5

1. Who invented the World Wide Web? Distinguish the web from the internet
2. Which protocol sends email? Which two receive it?
3. Distinguish DNS from DHCP
4. Give the port numbers for HTTP, HTTPS, FTP and SMTP
5. Distinguish a virus, a worm and a Trojan. What does a firewall do?

6. Session 6 — HTML ★★★★★

6.1 What HTML is

HTML — HyperText Markup Language — describes the **structure** of a web page.

> **MUST MASTER · HTML IS A MARKUP LANGUAGE, NOT A PROGRAMMING LANGUAGE**
>
> HTML has no variables, no loops and no conditional logic. It **marks up** content to describe what each part *is*. An option calling HTML a programming language is wrong.
>
> The division of labour is worth knowing: **HTML for structure, CSS for presentation, JavaScript for behaviour.**

6.2 Structure and tags

`` <html> <head> <title>Page title</title> </head> <body> content </body> </html> ``

- **Paired tags** have an opening and a closing form: `<p>` ... `</p>`
- **Empty tags** stand alone: `<br>` (line break), `<hr>` (horizontal rule), `<img>`
- **Attributes** appear inside the **opening** tag: `<img src="photo.jpg" alt="description">`
- Headings run **`<h1>` to `<h6>`**, with `<h1>` the largest
- Lists: `<ul>` unordered, `<ol>` ordered, with `<li>` for each item
- Tables: `<table>`, `<tr>` for a row, `<th>` for a header cell, `<td>` for a data cell

6.3 Links — the 2023 item

MUST MASTER · THE ANCHOR TAG, AND THE HASH FOR INTERNAL LINKS

All links use the **anchor tag** `<a>` with the `href` attribute. What changes is the value of `href`:

Kind of link	Syntax
To another **website**	`<a href="https://example.com">text</a>`
To another **page** in the same site	`<a href="page2.html">text</a>`
To a position WITHIN the same page	**`<a href="#sectionid">text</a>`**

The internal link is the one with the `#` (hash), and it points to the `id` of an element on the same page:

`` `<a href="#salary">Go to the salary section</a> ... <h2 id="salary">Salary</h2>` ``

Three facts that make this a reliable question:

1. **The attribute is `href`, never `src`.** `src` is for embedding a resource — an image or a script. `href` is for linking to one. Confusing them is the standard wrong option
2. **The `#` marks an in-page target.** Without it the browser looks for a file of that name
3. The target is identified by its `id` attribute

CHECK YOURSELF · SESSION 6

1. Is HTML a programming language? What do CSS and JavaScript do?
2. Distinguish paired from empty tags, with two examples of each
3. Where do attributes go?
4. Give the syntax for an internal link within the same page, and say what the target needs
5. `href` or `src` — which creates a link, and which embeds a resource?

7. Graded Practice — 25 Questions ★★★★★

7.1 Level 1 — Recognition

Q1. A computer network confined to a single building is a (a) WAN (b) LAN (c) MAN (d) PAN

Q2. The largest wide area network in existence is the (a) local area network (b) metropolitan area network (c) intranet (d) internet

Q3. In which topology is every node connected to every other node? (a) Bus (b) Star (c) Mesh (d) Ring

Q4. The number of layers in the OSI reference model is (a) seven (b) four (c) five (d) six

Q5. HTML stands for (a) HighText Machine Language (b) HyperText Markup Language (c) HyperText Machine Language (d) Hyperlink Text Marking Logic

Q6. An IPv4 address consists of (a) 8 bits (b) 16 bits (c) 64 bits (d) 32 bits

Q7. The protocol used for transferring files over the internet is (a) SMTP (b) HTTP (c) FTP (d) DNS

Q8. The World Wide Web was invented by (a) Tim Berners-Lee (b) Dennis Ritchie (c) Charles Babbage (d) Vint Cerf

Q9. DNS is used to (a) assign IP addresses dynamically (b) translate domain names into IP addresses (c) transfer files between hosts (d) encrypt network traffic

7.2 Level 2 — Understanding

Q10. In a full mesh network of n nodes, the number of links required is (a) n (b) n - 1 (c) n^2 (d) n(n - 1)/2

Q11. A full mesh network of 6 nodes requires how many links? (a) 6 (b) 12 (c) 15 (d) 30

Q12. Which one of the following is a **valid** IPv4 address? (a) 192.168.1.10 (b) 256.100.50.25 (c) 192.168.1 (d) 300.1.1.1

Q13. Each octet of an IPv4 address may take a value from (a) 1 to 256 (b) 0 to 255 (c) 0 to 127 (d) 1 to 255

Q14. A router operates at which layer of the OSI model? (a) Physical (b) Data link (c) Transport (d) Network

Q15. In HTML, a link to another position within the **same** page is created using (a) `<a src="page.html">` (b) `<link rel="internal">` (c) `<a href="#id">` (d) `<a name="page.html">`

Q16. Which transmission mode permits communication in both directions **simultaneously**? (a) Full duplex (b) Half duplex (c) Simplex (d) Unicast

Q17. Which device forwards data on the basis of MAC addresses? (a) Hub (b) Switch (c) Repeater (d) Modem

Q18. TCP differs from UDP in that TCP is (a) faster but unreliable (b) connectionless (c) used only for streaming media (d) connection-oriented and reliable

7.3 Level 3 — Recruitment Test standard

Q19. Which one of the following statements is **not** correct? (a) A mesh topology is the most reliable but the most expensive (b) In a star topology, failure of the central hub disables the whole network (c) In a bus topology, every node is connected to every other node (d) A hub operates at the physical layer

Q20. Consider the following statements:

1. An IPv4 address is written in dotted decimal notation as four octets.
2. Any octet with a value greater than 255 makes the address invalid.
3. IPv6 addresses are 64 bits long.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

Q21. Match List-I with List-II and select the correct answer using the code given below:

List-I	List-II
A. Hub	1. Physical layer
B. Switch	2. Data link layer
C. Router	3. Network layer
D. Gateway	4. Connects dissimilar networks

(a) A-2 B-1 C-3 D-4 (b) A-1 B-2 C-3 D-4 (c) A-1 B-2 C-4 D-3 (d) A-3 B-4 C-1 D-2

Q22. A full mesh network of 10 nodes requires (a) 90 links (b) 100 links (c) 10 links (d) 45 links

Q23. Which one of the following statements about HTML is **not** correct? (a) It is a markup language rather than a programming language (b) The anchor tag uses the href attribute to create a link (c) An internal link within a page is created using the src attribute (d) `<br>` is an empty tag

Q24. Consider the following statements:

1. Optical fibre offers higher bandwidth and immunity to electromagnetic interference.
2. A firewall filters traffic passing between networks.
3. SMTP is used for receiving electronic mail.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

Q25. The default port number for HTTP is (a) 21 (b) 80 (c) 443 (d) 25

8. Answers and Explanations

8.1 Level 1

Q	Ans	Explanation
1	b	A **LAN**. A MAN covers a city, a WAN a country or the world
2	d	The **internet**, the largest WAN
3	c	**Mesh.** This description is offered against bus, star and ring in almost every topology question
4	a	**Seven.** TCP/IP has four — the two counts are each other's distractors
5	b	**HyperText Markup Language**
6	d	**32 bits**, in four octets of 8 bits each. IPv6 is 128 bits
7	c	**FTP** — file transfer protocol
8	a	**Tim Berners-Lee**
9	b	**Translates domain names into IP addresses.** Option (a) is **DHCP**, which assigns them

8.2 Level 2

Q	Ans	Explanation
10	d	**n(n - 1)/2.** Each node links to n - 1 others, and each link is counted twice, so divide by 2
11	c	6 × 5 / 2 = **15**. Option (d), 30, is n(n - 1) without the division — the standard trap
12	a	**192.168.1.10.** (b) has an octet of 256, above the maximum; (c) has only three parts; (d) has 300
13	b	**0 to 255.** Option (a) is wrong at both ends — it excludes 0 and wrongly admits 256
14	d	The **network** layer, using IP addresses. A switch works at the data link layer using MAC addresses
15	c	`<a href="#id">` — the **hash** marks an in-page target, and the attribute is **href**, not src
16	a	**Full duplex** — the telephone. **Half duplex** is both ways one at a time, the walkie-talkie
17	b	A **switch**. A hub simply broadcasts to all ports without examining addresses
18	d	**Connection-oriented and reliable.** UDP is the connectionless, faster, unreliable one

8.3 Level 3

Q	Ans	Explanation
19	c	In a **bus** topology all nodes share a single **backbone cable**. It is the **mesh** topology in which every node connects to every other — so (c) has attached the mesh description to the bus. The other three are correct
20	a	**1 and 2 only.** Statement 3 is false — IPv6 is **128 bits**, not 64
21	b	**A-1 B-2 C-3 D-4** — hub physical, switch data link, router network, gateway joining dissimilar networks
22	d	10 × 9 / 2 = **45**. Option (a), 90, again omits the division by two
23	c	An internal link uses **href with a hash**, not **src**. `src` embeds a resource such as an image; `href` links to one. The other three statements are correct
24	a	**1 and 2 only.** Statement 3 is false — **SMTP sends** mail; **POP3 and IMAP receive** it
25	b	**80.** HTTPS is 443, FTP 21, SMTP 25

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

1. **The missing division.** Q11 and Q22 both offer n(n - 1) alongside the correct n(n - 1)/2. **If one option is exactly double another, the halved one is almost certainly right**
2. **The off-by-one range.** Q13. Eight bits give **256 values, 0 to 255**. The wrong option shifts the window by one at each end
3. **The attribute swap.** Q15 and Q23. **`href` links, `src` embeds.** One word decides both questions
4. **The transferred description.** Q19 takes the definition of mesh and attaches it to bus. Where a question lists several topologies, check that each description sits on the right one

9. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.7

BY AREA PAN » **LAN (building)** » **MAN (city)** » **WAN (country)** · **the internet is the largest WAN**

MEDIA optical fibre = highest bandwidth, immune to electromagnetic interference · simplex one way · **half duplex both ways one at a time (walkie-talkie)** · **full duplex both at once (telephone)**

TOPOLOGIES bus = one backbone, fails at the backbone · **star** = central hub, **fails at the hub** · **ring** = closed loop · **mesh = every node to every node**, most reliable, most expensive, **no single point of failure**

MESH FORMULA links = n(n - 1)/2 · ports per device = **n - 1** · **5 » 10** · **6 » 15** · **8 » 28** · **10 » 45** · the trap is **n(n - 1)** without halving

DEVICES hub broadcasts, physical layer · **switch uses MAC, data link** · **router uses IP, network** · gateway joins **dissimilar** networks · repeater regenerates · modem modulates and demodulates · NIC carries the **MAC address**

OSI = 7 LAYERS physical, data link, network, transport, session, presentation, application · **TCP/IP = 4 layers**

TCP vs UDP TCP **connection-oriented, reliable**, slower · UDP **connectionless**, faster, no guarantee

IPv4 32 bits, **four octets**, dotted decimal · **each octet 0 to 255** · 8 bits give **256 values but a maximum of 255** · invalid if any octet exceeds 255 **or** if there are not four parts · **IPv6 = 128 bits**, eight hexadecimal groups

CLASSES A 1–126 · B 128–191 · C 192–223 · D multicast · **127 = loopback (127.0.0.1)** · private: 10.x, 172.16–31.x, **192.168.x** · **MAC = 48 bits, hexadecimal, in hardware**

WEB Tim Berners-Lee · the web runs **on** the internet · URL = address

PROTOCOLS HTTP 80 · **HTTPS 443** · **FTP 21** · **SMTP 25 SENDS mail** · **POP3/IMAP RECEIVE** · **DNS 53 translates names to addresses** · **DHCP assigns addresses**

SECURITY virus attaches · worm spreads alone · Trojan disguises · ransomware encrypts for payment · phishing deceives · **firewall filters between networks** · cloud: IaaS, PaaS, SaaS

HTML a **MARKUP** language, **not** a programming language · **HTML structure, CSS presentation, JavaScript behaviour** · `<html>` `<head>` `<title>` `<body>` · paired vs **empty tags** (`<br>` `<hr>` `<img>`) · attributes go in the **opening** tag · headings **`<h1>` to `<h6>`** · `<ul>` `<ol>` `<li>` · `<table>` `<tr>` `<th>` `<td>`

LINKS all links use `<a>` **with** `href` · another site: full URL · another page: filename · **WITHIN the same page:** `href="#id"` **, pointing at an element's** `id` · `href` **LINKS,** `src` **EMBEDS**

10. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.7

MESH LINKS = n(n - 1)/2 · 5»10 · 6»15 · 10»45 · **if one option is double another, halve it**

MESH = EVERY NODE TO EVERY NODE · NO SINGLE POINT OF FAILURE

BUS FAILS AT THE BACKBONE · STAR FAILS AT THE HUB

IPv4 = 32 BITS, FOUR OCTETS, EACH 0–255 · 256 is INVALID · **IPv6 = 128 BITS**

HUB PHYSICAL · SWITCH MAC/DATA LINK · ROUTER IP/NETWORK

OSI 7 LAYERS · TCP/IP 4

TCP CONNECTION-ORIENTED AND RELIABLE · UDP CONNECTIONLESS AND FAST

SMTP SENDS · POP3 AND IMAP RECEIVE

DNS LOOKS UP · DHCP HANDS OUT

HTTP 80 · HTTPS 443 · FTP 21 · SMTP 25 · DNS 53

OPTICAL FIBRE: HIGHEST BANDWIDTH, IMMUNE TO EMI

HALF DUPLEX = WALKIE-TALKIE · FULL DUPLEX = TELEPHONE

HTML IS MARKUP, NOT PROGRAMMING

INTERNAL LINK = `<a href="#id">` **· href LINKS, src EMBEDS**

ONE LINE FOR THE INTERVIEW A mesh network needs n(n - 1)/2 links because every node must reach every other and each link serves both ends — which is exactly why mesh is unbeatable for reliability and unaffordable at scale.

ACTION · MODULE 9 IS COMPLETE

Seven chapters, F9.1 to F9.7, and **all twenty of the 2023 General Science and Computer Applications items** recorded in Blueprint §4.4 are now covered.

Next: Module 8 — General English, Quantitative Aptitude and Reasoning. It is **80 planning marks, the largest single block in the paper**, and the Blueprint requires it to run as **daily practice** rather than as a block reached in sequence. The statistics in F6.8 and the mensuration and physics arithmetic in F9.1 are already part of it.

Before you leave Module 9, state without looking: the mesh link formula; the valid range of an IPv4 octet; which layer a router works at; which protocol sends mail; and how an internal HTML link is written.